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(54) BEARING DEVICE WITH INTEGRATED ELECTRICAL INSULATION, NOTABLY FOR AN ELECTRIC MOTOR OR MACHINE

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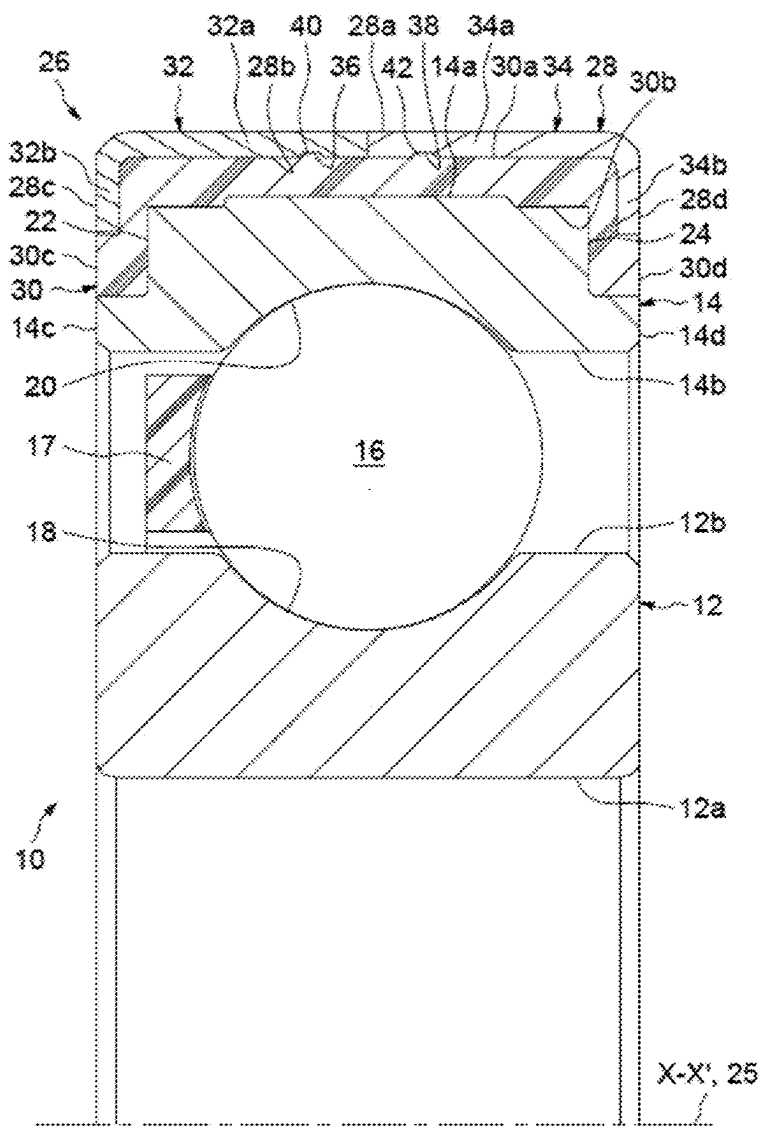
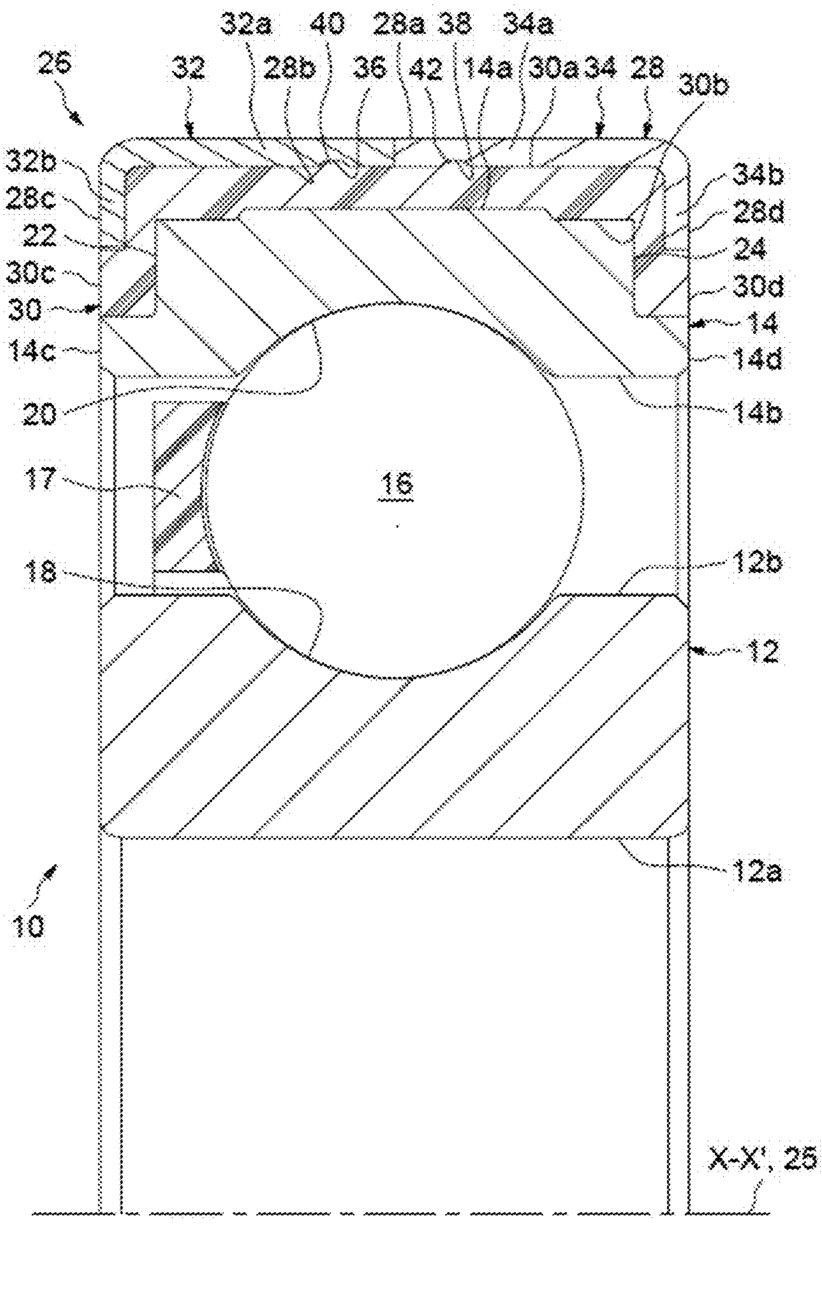
(57) **ABSTRACT**

A bearing device includes a bearing having first and second rings, a bushing and an electrically insulating insert overmolded between and connecting the bushing and a cylindrical surface of the second ring. The bushing is formed from two parts each having a cylindrical portion and a radial collar, and the insert is overmolded onto the cylindrical portions and the collars. Also, a surface of the first cylindrical portion and a surface of the second cylindrical portion together delimit a radially inner surface of the bushing or a radially outer surface of the bushing.





32a
40
28a
38
34a
26
32
28b
36
42
14a
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28
30b
32b
28c
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30c
30
14c
20
17
18
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28d
24
30d
14
14d
14b
16
12b
12
12a
X-X', 25

FIG. 1

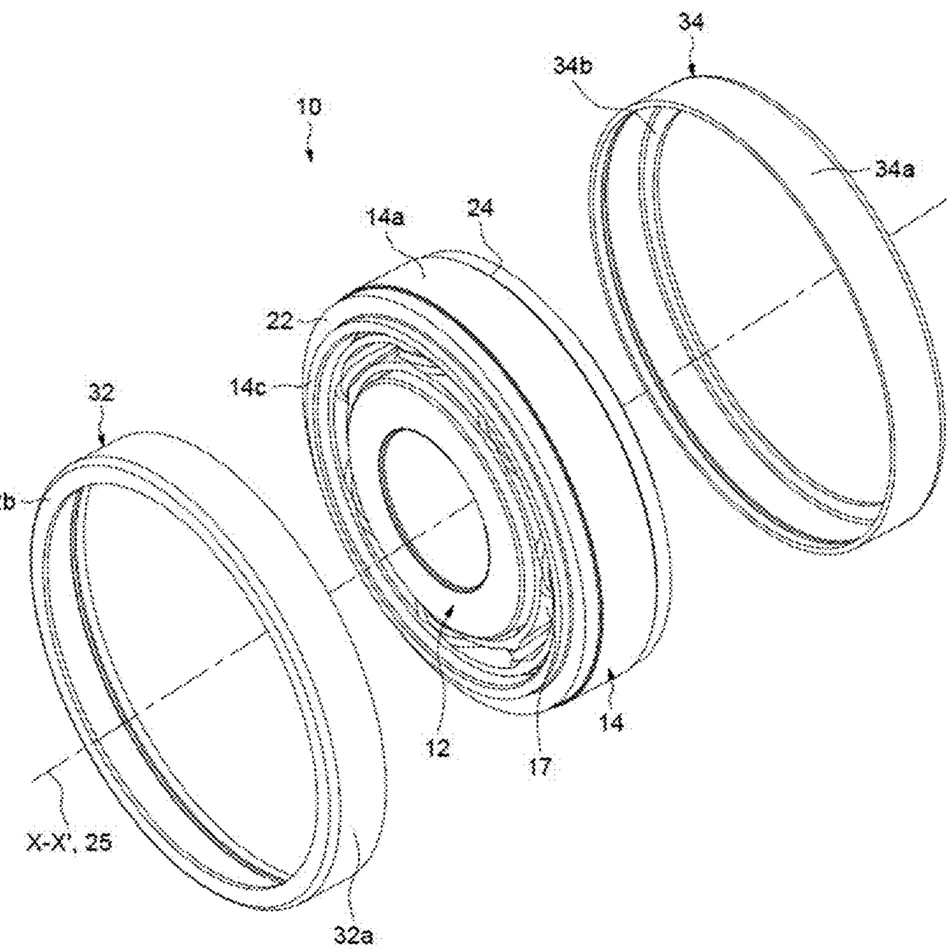

FIG. 2

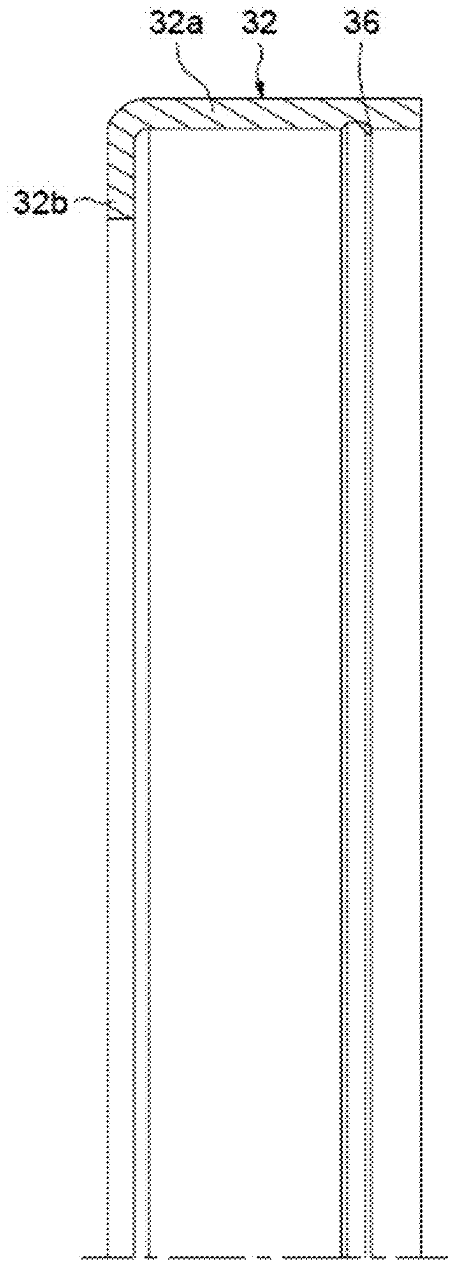

FIG. 3

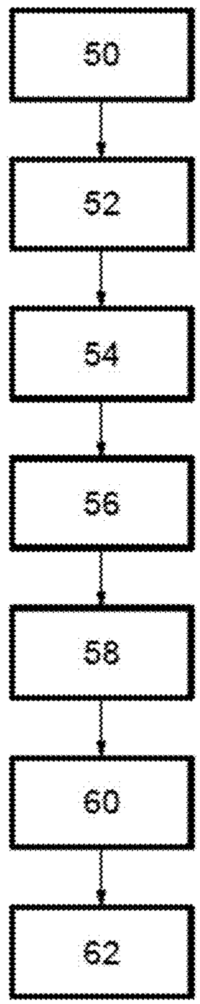

FIG. 4

BEARING DEVICE WITH INTEGRATED ELECTRICAL INSULATION, NOTABLY FOR AN ELECTRIC MOTOR OR MACHINE

CROSS-REFERENCE

[0001] This application claims priority to French patent application no. 2402410 filed on Mar. 11, 2024, the contents of which are fully incorporated herein by reference.

TECHNOLOGICAL FIELD

[0002] The present disclosure relates to the field of bearings that are particularly used in electric motors, electric machines and associated equipment.

BACKGROUND

[0003] In an electric motor or machine, at least one rolling bearing is mounted between the housing of the electric motor or machine and the rotary shaft in order to support this shaft. During operation, when the shaft is rotating, a difference in electric potential can occur between the shaft and the housing of the motor or of the electric machine, generating an electric current between the inner ring of the rolling bearing, which is rigidly connected to the shaft, and the outer ring, which is rigidly connected to the housing. The electric current flowing through the components of the rolling bearing can damage these components, notably the rolling elements and raceways provided on the inner and outer rings. Electrical discharges can also generate vibrations.

[0004] In order to overcome these disadvantages, a known solution involves replacing the rolling elements of the bearing, which are made from the same steel as the inner and outer rings, with rolling elements made of ceramic. This is generally referred to as a hybrid rolling bearing. However, such a hybrid rolling bearing is relatively expensive.

[0005] In order to overcome the aforementioned disadvantages, another known solution involves equipping the outer ring of the rolling bearing with an insulating sleeve provided with a bushing and an insulating insert made of electrically insulating material and radially interposed between the outer ring and the bushing. In order to attach the insulating insert to the outer ring and to the bushing without any additional elements or specific machining on the outer ring, the insulating insert can be overmolded. However, with such a solution, relative uncoupling of the insulating insert and of the bushing can sometimes occur during operation.

SUMMARY

[0006] Therefore, an aspect of the present disclosure is to overcome the aforementioned disadvantages by providing a bearing device with a simple and economical design. Another aspect of the disclosure is a bearing device comprising a bearing having a first ring and a second ring that are able to rotate relative to one another.

[0007] The device further comprises at least one insulating sleeve mounted on the second ring of the bearing. The insulating sleeve is provided with a bushing and an insulating insert radially interposed between the second ring of the bearing and the bushing. The insulating insert is made of an electrically insulating material.

[0008] The bushing has an outer surface and an inner surface, opposite the outer surface, which delimit the radial thickness of the bushing. The insulating insert is overmolded onto the second ring of the bearing and at least onto one of the outer and inner surfaces of the bushing.

[0009] According to a general feature, the bushing is produced in at least two distinct parts, each part comprising an axial portion and a radial collar extending radially from the axial portion. According to another general feature, the axial portions of the two parts at least partly jointly delimit the surface of the bushing onto which the insulating insert is overmolded. According to yet another general feature, the insulating insert is also overmolded onto an inner face of the radial collar of each of the two parts of the bushing.

[0010] Producing the two parts of the bushing with radial collars allows a good connection to be obtained with the insulating insert. The risk of any relative movements between the insulating insert and the bushing in the axial direction is avoided, notably during temperature variations.

[0011] As used herein, "axial direction" is understood to mean the direction parallel to the axis of the bearing device.

[0012] Furthermore, compared with a one-piece construction of the bushing equipped with collars, producing the bushing in at least two distinct parts facilitates the installation of the second ring inside the mold provided for overmolding the insulating insert.

[0013] This provides a bearing device with integrated electrical insulation that is economical compared with conventional hybrid rolling bearings and is easy to manufacture.

[0014] Preferably, the radial collar of at least one of the two parts of the bushing radially extends beyond an outer or inner surface of the second ring onto which the insulating insert is overmolded.

[0015] With such an arrangement, the part of the insulating insert that is axially located between the second ring and this radial collar of the bushing does not experience shearing stresses when high axial loads are applied to the device mounted inside the housing of the associated electric motor or machine with this collar in abutment against a shoulder of the housing. Indeed, in this case, compression stresses are applied to this part of the insulating insert. This increases the reliability of the device.

[0016] Advantageously, the radial collar of each of the two parts of the bushing radially extends beyond the outer or inner surface of the second ring onto which the insulating insert is overmolded. As a variant, it is nevertheless possible for at least one of the two radial collars or both collars to be flush with the outer or inner surface of the second ring or to remain radially set back. In one embodiment, the two parts of the bushing are symmetrical relative to a radial midplane of the device. This reduces the manufacturing cost of the device.

[0017] According to a particular design, the axial portions of the two parts of the bushing are axially in contact against each other and jointly delimit the whole of the surface of the bushing onto which the insulating insert is overmolded.

[0018] According to another design, the axial portions of the two parts of the bushing are axially spaced apart from each other. In this case, the bushing can further comprise an additional ring axially interposed between the axial portions of the two parts and jointly delimiting, with the axial portions of the two parts of the bushing, the surface of the bushing onto which the insulating insert is overmolded.

[0019] According to a particular design, the surface of the bushing is provided with at least one groove extending in the circumferential direction and inside which a rib of matching shape extends for attaching the insulating insert. This further increases the axial attachment of the insulating insert on the bushing.

[0020] As used herein, "circumferential direction" is understood to mean the direction that is perpendicular both to the axial direction and to a radius of the bearing device, in other words, tangent to a circle whose center is on the axis of the bearing device.

[0021] Each axial portion of the two parts of the bushing can be provided with at least one groove extending in the circumferential direction and inside which a rib of matching shape extends for attaching the insulating insert.

[0022] The bushing can be provided with two front faces delimiting its axial length.

[0023] According to a first design, the radial collar of at least one of the two parts is axially flush with one of the front faces of the bushing.

[0024] According to a second design, the radial collar of each of the two parts of the bushing is axially flush with one of the front faces of the bushing.

[0025] Alternatively, one or each of the radial collars can be axially inwardly or outwardly offset relative to the associated front surface of the bushing.

[0026] If the insulating insert is made of synthetic or elastomer material, it makes the device insensitive to temperature variations.

[0027] In a particular embodiment, the bushing is made of metal material. The bushing thus can be easily machined to a predetermined radial tolerance.

[0028] In one embodiment, the insulating insert covers the whole of the surface of the bushing. In this case, the insulating insert completely covers the surface of the bushing in the axial direction and in the circumferential direction.

[0029] According to a first design, the bushing delimits the outer surface of the device. In this case, the second ring is the outer ring of the bearing. According to a second alternative design, the bushing delimits the inner surface of the device. In this case, the second ring is the inner ring of the bearing.

[0030] According to an embodiment, a bearing device comprises a bearing including a first ring and a second ring configured to rotate relative to each other about a central axis, the second ring having a first cylindrical surface and a second cylindrical surface radially spaced from the first cylindrical surface. The device also include a bushing and an electrically insulating insert overmolded between and connecting the bushing and the second cylindrical surface of the second ring. The bushing comprises a first part having an axially extending first cylindrical portion and a first annular collar extending radially from a first axial end of the first cylindrical portion and a second part having an axially extending second cylindrical portion and a second annular collar extending radially from a first axial end of the second cylindrical portion. A first surface of the first cylindrical portion faces the second ring and a first surface of the second cylindrical portion faces the second ring and a first surface of the first collar faces a first surface of the second collar. The electrically insulating insert is overmolded onto the first surface of the first cylindrical portion and onto the first surface of the second cylindrical portion and onto the first surface of the first collar and onto the first surface of the second collar. Also a second surface of the first cylindrical portion is radially spaced from the first surface of the first cylindrical portion and a second surface of the second cylindrical portion is radially spaced from the first surface of the second cylindrical portion, and the second surface of the first cylindrical portion and the second surface of the second cylindrical portion together delimit a radially inner surface of the bushing or a radially outer surface of the bushing.

[0031] In a particular embodiment, the bearing comprises at least one row of rolling elements disposed between raceways of the first and second rings. The rolling elements can be made of metal material.

[0032] The disclosure also relates to an electric motor comprising a housing, a shaft and at least one bearing device as defined above and radially mounted between the housing and the shaft.

[0033] The disclosure also relates to a method for manufacturing a bearing device as defined above, that includes a step of mounting one of the parts of the bushing in the bottom of a manufacturing mold, a step of installing the second ring inside the manufacturing mold, a step of mounting the other one of the parts of the bushing inside the manufacturing mold, a step of overmolding the insulating insert onto the second ring and at least onto the surface of the bushing, and a step of assembling, with the first ring of the bearing, the assembly formed by the second ring, the parts of the bushing and the insulating insert.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] The present invention will be better understood with reference to the detailed description of an embodiment, which is provided by way of a non-limiting example and is illustrated by the appended drawings, in which:

[0035] FIG. **1** is a half-view of an axial section of a bearing device according to one embodiment of the disclosure.

[0036] FIG. **2** is a partial exploded perspective view of the bearing device of FIG. **1** with the insulating insert removed for illustration purposes.

[0037] FIG. **3** is a half-view of an axial section of one of the parts of a bushing of the bearing device of FIG. **1**.

[0038] FIG. **4** is a flowchart illustrating the method for manufacturing the bearing device of FIG. **1**.

DETAILED DESCRIPTION

[0039] The bearing device illustrated in FIG. **1** comprises a bearing **10** provided with a first ring **12** and a second ring **14** that are configured to rotate relative to each other about the X-X' axis of the bearing. In the illustrated embodiment, the first ring **12** is the inner ring of the bearing and the second ring **14** is the outer ring.

[0040] As will be described in further detail hereafter, the bearing device is designed so that it does not conduct electric currents. The bearing device has integrated electric insulation.

[0041] The inner **12** and outer **14** rings of the bearing are concentric and extend axially along the X-X' axis of the bearing. The inner **12** and outer **14** rings are made of steel. The rings are solid type rings.

[0042] In the illustrated embodiment, the bearing **10** also comprises a row of rolling elements **16**, in this case balls, radially interposed between the inner **12** and outer **14** rings. The rolling elements **16** are made of steel. The bearing **10** also comprises a cage **17** for maintaining the even circumferential spacing of the rolling elements **16**. The bearing **10** also can be equipped with sealing seals or flanges.

[0043] The inner ring **12** comprises a cylindrical bore **12***a*, a cylindrical axial outer surface **12***b* radially opposite the bore, and two opposite radial front faces (not referenced) axially delimiting the bore and the outer surface. The bore **12***a* and the outer surface **12***b* delimit the radial thickness of the inner ring **12**. The bore **12***a* forms the inner surface of the inner ring. The inner ring **12** also comprises an inner raceway **18** for the rolling elements **16** that is formed on the outer surface **12***b*. The raceway **18** is directed radially outwards.

[0044] The outer ring **14** comprises a cylindrical axial outer surface **14***a*, a cylindrical bore **14***b* radially opposite the outer surface **14***a*, and two opposite radial front faces **14***c*, **14***d* axially delimiting the bore. The outer surface **14***a* and the bore **14***b* delimit the radial thickness of the outer ring **14**. In the illustrated embodiment, the outer surface **14***a* of the ring has portions having two distinct diameters. Alternatively, the outer surface **14***a* could have a single diameter. The outer ring **14** further comprises an outer raceway **20** for the rolling elements **16**, which is formed on the bore **14***b*. The raceway **20** is directed radially inwards.

[0045] In the illustrated embodiment, a groove **22** is provided on the front face **14***c* of the outer ring. In other words, a diameter of the outer ring at the front face **14***c* is smaller than a diameter of the outer ring at a distance from the front face **14***c*. The groove **22** is axially oriented towards the outside of the outer ring. The groove **22** has a bottom that is axially offset towards the inside of the ring relative to the front face **14***c*. The bottom of the groove **22** forms a shoulder. The bottom of the groove **22** in this case extends radially for ease of manufacturing. The groove **22** is annular in this case.

[0046] Similarly, a groove **24** is provided on the front face **14***d* of the outer ring. In other words, a diameter of the outer ring at the front face **14***d* is smaller than a diameter of the outer ring at a distance from the front face **14***d*. The groove **24** is axially oriented towards the outside of the outer ring. The groove **24** has a bottom that is axially offset towards the inside of the ring relative to the front face **14***d*. The bottom of the groove **24** forms a shoulder. The bottom of the groove **24** extends radially in this case. The groove **24** is annular in this case. The grooves **22**, **24** are symmetrical with each other relative to a radial midplane of the outer ring. The grooves **22**, **24** axially delimit the outer surface **14***a*.

[0047] The bearing device also comprises an electrically insulating sleeve **26** mounted on the outer ring **14**. The insulating sleeve **26** is mounted on the outer surface **14***a* of the outer ring **14**. The insulating sleeve **26** is rigidly connected to the outer ring **14**. The insulating sleeve **26** comprises a bushing **28** and an insulating insert **30** radially interposed between the outer ring **14** and the bushing **28**. The insulating insert **30** is overmolded onto the outer ring **14** and the bushing **28**.

[0048] The bushing **28** is annular. The bushing **28** is made up of two distinct parts: a first part **32** and a second part **34**. These two separate parts **32**, **34** form half-flanges that in this case are axially in abutment against each other. In the illustrated embodiment, the parts **32**, **34** of the bushing are identical and symmetrical relative to a radial midplane of the device in order to reduce the manufacturing costs. Alternatively, it is of course possible for non-symmetrical parts **32**, **34** to be provided. In another variant, it is possible for the bushing **28** to be made up of more than two parts. Preferably, the parts **32**, **34** of the bushing **28** are made of steel. The parts **32**, **34** advantageously can be obtained from a metal sheet by cutting, stamping and rolling. Alternatively, the parts **32**, **34** can be obtained from a tube or from forged/rolled blanks, or even from sintering and stamping.

[0049] The first part **32** of the bushing includes an axially extending first cylindrical portion **32***a* and a first annular collar **32***b* extending radially inward from an axial end of the first cylindrical portion. The second part **34** of the bushing includes an axially extending second cylindrical portion **34***a* and a second annular collar **34***b* extending radially inward from an axial end of the second cylindrical portion. The axially extending portions **32***a*, **34***a* are axially in abutment against each other. The annular collars **32***b*, **34***b* extend inward from axially opposite ends of the axially extending portion **32***a*, **34***a* axially located on the outside of the device. In the illustrated embodiment, the collars **32***b*, **34***b* are annular. Alternatively, at least one of the collars **32***b*, **34***b* could be in the form of sectors spaced apart from each other in the circumferential direction.

[0050] The bushing **28** comprises a cylindrical axial outer surface **28***a*, and a cylindrical bore **28***b* radially opposite the outer surface **28***a*, the axis **25** of which is coaxial with the X-X' axis. The bore **28***b* forms the inner surface of the bushing **28**. The axial portions **32***a*, **34***a* of the parts of the bushing jointly delimit the outer surface **28***a*. Similarly, the axial portions **32***a*, **34***a* of the parts jointly delimit the bore **28***b*. The outer surface **28***a* and the bore **28***b* delimit the radial thickness of the bushing **28**. The outer surface **28***a* of the bushing forms the outer surface of the bearing device **10**. In other words, the outer surface **28***a* defines the outer diameter of the bearing device **10**.

[0051] The bushing **28** also comprises two opposite radial front faces **28***c*, **28***d* axially delimiting the outer surface **28***a*. The front faces **28***c*, **28***d* delimit the axial length of the bushing. The front face **28***c* is delimited by the radial collar **32***b*, and the front face **28***d* is delimited by the radial collar **34***b*. More specifically, the front face **28***c* is delimited by the outer face of the radial collar **32***b*, and the front face **28***d* is delimited by the outer face of the radial collar **34***b*.

[0052] In the illustrated embodiment, the front faces **28***c*, **28***d* of the bushing are respectively coplanar with the front faces **14***c*, **14***d* of the outer ring. Alternatively, other arrangements can be provided. For example, the bushing **28** could have a smaller or greater axial dimension and could remain axially set back from the faces **14***c*, **14***d* of the outer ring, or could project from the faces.

[0053] In the illustrated embodiment, the radial collars **32***b*, **34***b* of the parts of the bushing radially extend beyond the outer surface **14***a* of the outer ring, i.e., radially project inwards relative to the outer surface **14***a*. In other words, the free ends of the radial collars **32***b*, **34***b* are radially offset inwards relative to the outer surface **14***a* of the outer ring. The radial collars **32***b*, **34***b* partly extend into the grooves **22**, **24** of the outer ring. The radial collars **32***b*, **34***b* remain at a distance from the inner ring **12**.

[0054] The bore of the axial portion **32***a*, **34***a* of each part **32**, **34** of the bushing is provided with a groove **36**, **38** that circumferentially extends around the axis **25** of the bore of the bushing. Each groove **36**, **38** is radially oriented towards the outer ring **14**, i.e., radially inwards.

[0055] In the illustrated embodiment, each groove **36**, **38** is annular. Alternatively, at least one of the two grooves **36**, **38** may not extend over 360°, or even may be formed by a succession of circumferentially extending turns that are spaced apart from one another in the circumferential direction.

[0056] Each groove **36**, **38** is delimited in the axial direction by two opposing lateral flanks that have a straight profile as an axial section and are connected together by an axial bottom. Alternatively, other shapes can be provided, for example, grooves that in this case, as a cross section, assume the shape of an inwardly oriented arc of a circle. In another variant, the bushing **28** also can be devoid of grooves **36**, **38**.

[0057] The insulating insert **30** is made of electrically insulating material. The insulating insert **30** can be made, for example, of synthetic material, such as PEEK or PA46, or it even can be made of elastomer material, such as rubber, for example.

[0058] The insulating insert **30** is radially interposed between the outer surface **14***a* of the outer ring and the bore **28***b* of the bushing. The insulating insert **30** covers the outer surface **14***a* of the outer ring. In this case, the insulating insert **30** completely covers the outer surface **14***a* in the axial and circumferential directions. The insulating insert **30** also covers the grooves **22**, **24** of the outer ring

[0059] The insulating insert **30** also covers the bore **28***b* of the bushing. The insulating insert **30** in this case also completely covers the bore **28***b* in the axial and circumferential directions. The insulating insert **30** covers the bore of the axial portion **32***a*, **34***a* of each part **32**, **34** of the bushing.

[0060] The insulating insert **30** also covers the inner face of the radial collar **32***b*, **34***b* of each part **32**, **34** of the bushing. The inner face and the outer face axially opposite the inner face of each radial collar **32***b* and **34***b* delimit the axial thickness of the collar. For each radial collar **32***b* and **34***b*, the inner face is axially oriented towards the inside of the device, and the outer face is axially oriented towards the outside of the device. The insulating insert **30** also covers the free end of the radial collar **32***b*, **34***b* of each part **32**, **34** of the bushing.

[0061] The insulating insert **30** is annular. The insulating insert **30** extends axially. The insulating insert **30** comprises a cylindrical axial outer surface **30***a*, a cylindrical bore **30***b* radially opposite the outer surface **30***a*, and two opposite radial front faces **30***c*, **30***d* axially delimiting the bore and the outer surface. The radial front faces **30***c*, **30***d* axially delimit the insulating insert **30**. The outer surface **30***a* and the bore **30***b* delimit the radial thickness of the insulating insert **30**. The outer surface **30***a* is in radial contact with the bore **28***b* of the bushing. The outer surface **30***a* is also in radial contact with the free end of the radial collar **32***b*, **34***b* of each part **32**, **34** of the bushing. The outer surface **30***a* assumes a stepped shape. The bore **30***b* is in radial contact with the outer surface **14***a* of the outer ring and with the grooves **22**, **24**. The bore **30***b* has a stepped shape.

[0062] In the illustrated embodiment, the faces **14***c*, **30***c*, **28***c* and **14***d*, **30***d*, **28***d* of the outer ring, the insulating insert and the bushing are respectively coplanar.

[0063] Alternatively, other arrangements can be provided. For example, the insulating insert **30** could have a limited axial dimension and could remain axially set back from the faces **14***c*, **14***d* of the outer ring. Alternatively, the insulating insert **30** could have a greater axial dimension and axially project from the faces **14***c*, **14***d* of the outer ring. In this case, the insulating insert **30** can at least partly cover these faces **14***c*, **14***d*. As a variant, the insulating insert **30** could at least partly cover the faces **28***c*, **28***d* of the bushing.

[0064] In another alternative, or in combination, the bushing **28** could axially project from the insulating insert **30** relative to the faces **30***c* and **30***d*, or could remain axially set back from these faces.

[0065] The insulating insert **30** also comprises two ribs **40**, **42** extending radially outwards from the outer surface **30***a* and each accommodated inside one of the grooves **36**, **38** of the bushing. The rib **40**, **42** assumes a shape that matches the associated groove **36**, **38**. Each rib **40**, **42** projects from the outer surface **30***a* of the insulating insert. Each rib **40**, **42** is formed on the outer surface **30***a* when overmolding the insulating insert **30**.

[0066] The bearing device is manufactured as follows.

[0067] In a first step **50**, the first part **32** of the bushing is mounted inside a mold, which is provided for overmolding the insulating insert **30**.

[0068] In a second successive step **52**, the outer ring **14** equipped with the grooves **22**, **24** is installed inside the mold.

[0069] Then, in a third step **54**, the second part **34** of the bushing is mounted inside the mold, axially in abutment against the first part **32**. In this position mounted inside the mold, the first part **32** and the second part **34** of the bushing are radially at a distance from the outer ring **14**.

[0070] Next, in a fourth successive step **56**, the insulating insert **30** is overmolded both onto the outer ring **14** and onto the first and second parts **32**, **34** of the bushing **28**. The ribs **40**, **42** in the insulating insert are formed during this step.

[0071] In a fifth successive step **58**, the unitary assembly formed by the outer ring **14**, the first and second parts **32**, **34** of the bushing **28** and the insulating insert **30** is extracted from the mold.

[0072] Then, in a sixth successive step **60**, the front faces **14***c*, **14***d* and the faces **28***c*, **28***d* of the bushing are ground. Given the presence of the collars **32***b* and **34***b* of the bushing, the grinding operation is mainly carried out on the bushing and the outer ring, and not on the insulating insert **30**. During this step, the outer surface **28***a* of the bushing and the raceway **20** of the outer ring also can be ground.

[0073] Then, in a seventh step **62**, the unitary assembly formed by the outer ring **14**, the first and second parts **32**, **34** of the bushing **28** and the insulating insert **30** is assembled with the row of rolling elements **16**, the cage **17** and the inner ring **12**.

[0074] In the illustrated embodiments, the first ring **12** of the bearing is the inner ring and the second ring **14**, onto which the insulating insert **30** is overmolded, is the outer ring.

[0075] Alternatively, a reverse arrangement can be provided whereby the second ring **14**, onto which the insulating insert **30** is overmolded, is the inner ring. In this case, the insulating sleeve is located in the bore **12***a* of the inner ring. The insulating insert is then radially interposed between the bore **12***a* of the inner ring and the outer surface of the bushing. The insulating insert is overmolded onto the inner ring and at least onto the outer surface of the bushing. The bore of the bushing delimits the bore of the bearing device.

[0076] In the described embodiments, the bearing of the device is provided with a single row of rolling elements. As a variant, the bearing can be provided with several rows of rolling elements. In addition, the rolling bearing can include types of rolling elements other than balls, for example, rollers. In another variant, the bearing can be a slider bearing devoid of rolling elements.

[0077] Representative, non-limiting examples of the present invention were described above in detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the invention. Furthermore, each of the additional features and teachings disclosed above may be utilized separately or in conjunction with other features and teachings to provide improved insulated bearings.

[0078] Moreover, combinations of features and steps disclosed in the above detailed description may not be necessary to practice the invention in the broadest sense, and are instead taught merely to particularly describe representative examples of the invention. Furthermore, various features of the above-described representative examples, as well as the various independent and dependent claims below, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

[0079] All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.

What is claimed is:

1. A bearing device comprising:

a bearing including a first ring and a second ring configured to rotate relative to each other about a central axis, the second ring having a first cylindrical surface and a second cylindrical surface radially spaced from the first cylindrical surface,

a bushing, and

an electrically insulating insert overmolded between and connecting the bushing and the second cylindrical surface of the second ring,

wherein the bushing comprises a first part having an axially extending first cylindrical portion and a first annular collar extending radially from a first axial end of the first cylindrical portion, and a second part having an axially extending second cylindrical portion and a second annular collar extending radially from a first axial end of the second cylindrical portion,

wherein a first surface of the first cylindrical portion faces the second ring and a first surface of the second cylindrical portion faces the second ring and wherein a first surface of the first collar faces a first surface of the second collar,

wherein the electrically insulating insert is overmolded onto the first surface of the first cylindrical portion and onto the first surface of the second cylindrical portion and onto the first surface of the first collar and onto the first surface of the second collar, and

wherein a second surface of the first cylindrical portion is radially spaced from the first surface of the first cylindrical portion and a second surface of the second cylindrical portion is radially spaced from the first surface of the second cylindrical portion, and

wherein the second surface of the first cylindrical portion and the second surface of the second cylindrical portion together delimit a radially inner surface of the bushing or a radially outer surface of the bushing.

2. The bearing device according to claim **1**,

wherein a second axial end of the first cylindrical portion contacts a second axial end of the second cylindrical portion at a joint.

3. The bearing device according to claim **1**,

wherein the bushing and the second ring are configured such that an imaginary cylinder centered on the central axis intersects the first annular collar and the second ring.

4. The bearing device according to claim **1**,

wherein the bushing and the second ring are configured such that an imaginary cylinder centered on the central axis intersects the first annular collar and the second annular collar and the second ring.

5. The bearing device according to claim **1**,

wherein the first part and the second part are symmetrical relative to a radial midplane of the second ring.

6. The bearing device according to claim **1**,

wherein the bushing includes at least one radially facing groove, and

wherein a rib of the electrically insulating insert extends into the at least one radially facing groove.

7. The bearing device according to claim **1**,

wherein the first surface of the first cylindrical portion includes at least one radially facing groove and the first portion of the second cylindrical portion includes at least one radially facing groove, and

wherein at least one first rib of the electrically insulating insert extends into the at least one radially facing groove of the first cylindrical portion and at least one second rib of the electrically insulating insert extends into the at least one radially facing groove of the second cylindrical portion.

8. The bearing device according to claim **1**,

wherein a second surface of the first collar axially opposite the first surface of the first collar delimits a first axial end of the bushing and a second surface of the second collar axially opposite the first surface of the second collar axially delimits a second axial end of the bushing.

9. The bearing device according to claim **1**,

wherein a second axial end of the first cylindrical portion contacts a second axial end of the second cylindrical portion at a joint,

wherein the bushing and the second ring are configured such that an imaginary cylinder centered on the central axis intersects the first annular collar and the second ring,

wherein the first part and the second part are symmetrical relative to a radial midplane of the second ring,

wherein the bushing includes at least one radially facing groove,

wherein a rib of the electrically insulating insert extends into the at least one radially facing groove, and

wherein a second surface of the first collar axially opposite the first surface of the first collar delimits a first axial end of the bushing and a second surface of the second collar axially opposite the first surface of the second collar axially delimits a second axial end of the bushing.

10. A method for manufacturing a bearing device according to claim **1** comprising:

a step of mounting one of the parts of the bushing in the bottom of a manufacturing mold;

a step of installing the second ring inside the manufacturing mold;

a step of mounting the other one of the parts of the bushing inside the manufacturing mold;

a step of overmolding the insulating insert onto the second ring and at least onto the surface of the bushing; and

a step of assembling, with the first ring of the bearing, the assembly formed by the second ring, the parts of the bushing and the insulating insert.

11. An electric motor comprising:

a housing,

a shaft, and

at least one bearing device according to claim **1** radially mounted between the housing and the shaft.

12. An electric motor comprising:

a housing,

a shaft, and

at least one bearing device according to claim **9** radially mounted between the housing and the shaft.

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